FRB
FONDATION POUR LA RECHERCHE SUR LA BIODIVERSITÉ

CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE SUR LA BIODIVERSITÉ

INTRODUCTION AUX MÉTA-ANALYSES ET REVUES SYSTÉMATIQUES

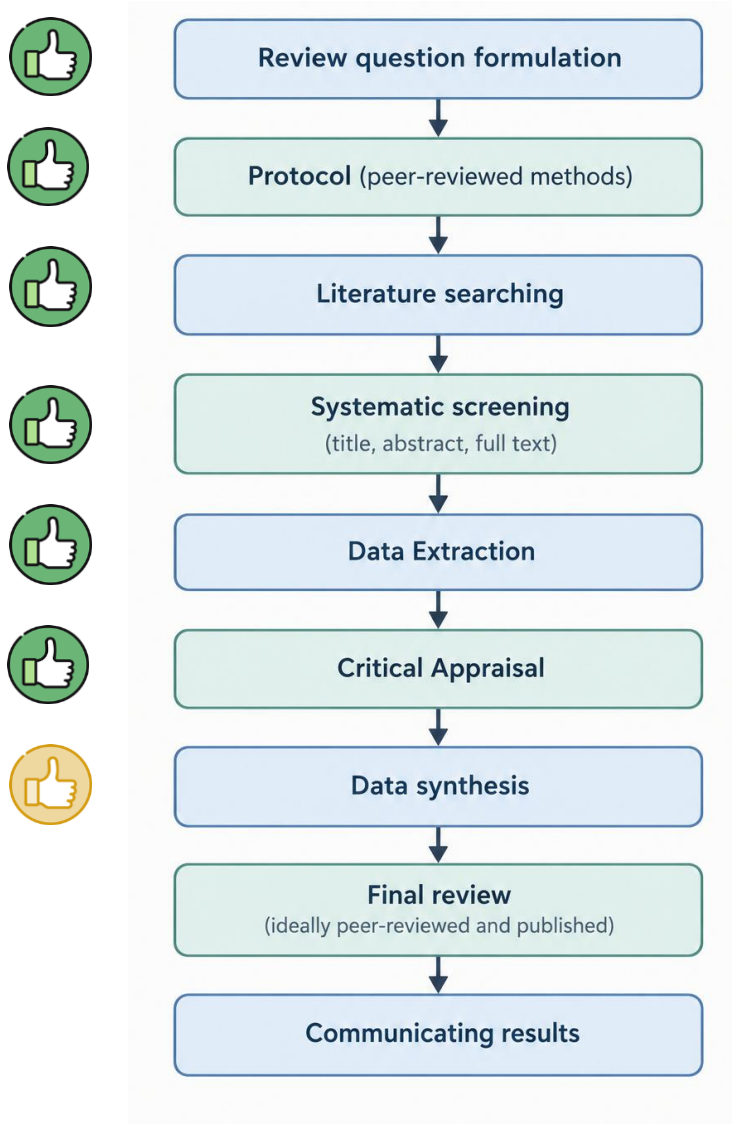
Quantitative data extraction –
Importance of correctly formatting datasets

Joseph LANGRIDGE – Chargé de mission scientifique « Revues systématiques et synthèses »

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Recap: steps taken



Extracting quantitative data: what does it entail?

What quantitative data is needed to calculate **effect sizes** ?

You need: **mean**, **sample size**, **sd/se/95% CI**.

+ extraction of variables that could explain the heterogeneity of effect sizes (***effect modifiers***)

where to extract data from ?

- Text
- Tables
- Figures
- supp. mats.
- Calculations may be needed (keep a record)

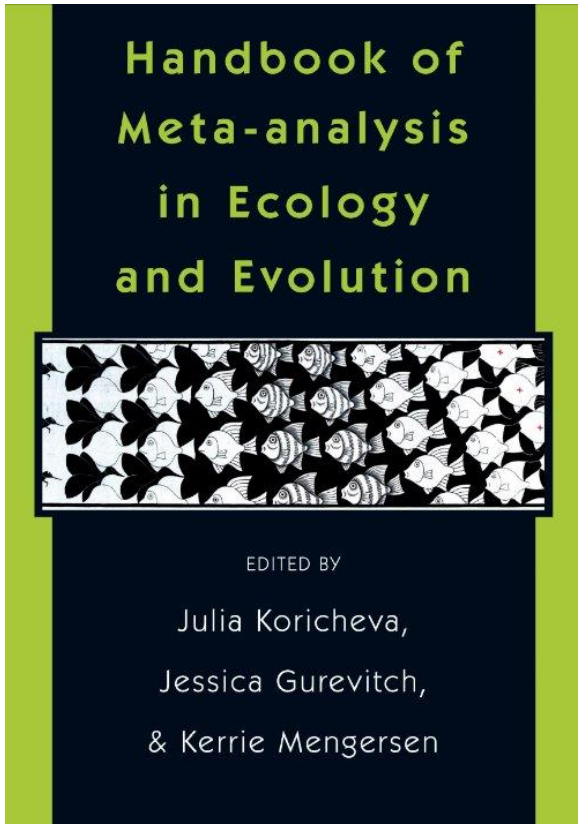
Extracting data: what does it entail?

! BE AWARE !

- Data extraction is **time-consuming**: clearly define the extraction sheet and the *effect modifiers* to be extracted.

- **<u>Important to run pilot tests</u>** using the drafted extraction sheet.
  - Do it on a small sample of articles to check that it **matches the content** of the studies.

- Decide what to do in case of **missing information** ("missing data", contact the authors, imputation)



Format your database accordingly

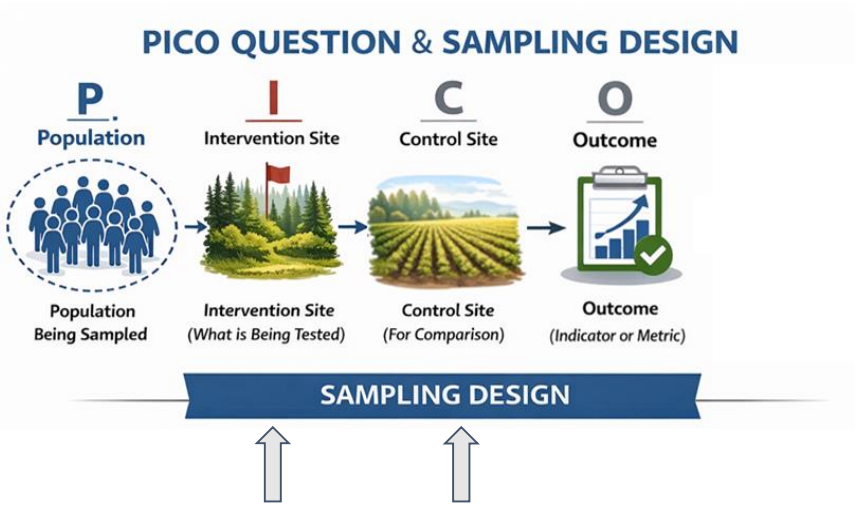
Define what a comparative study is (follow your PICO)

Example 1, Langridge et al. 2023:

- Population 1 × intervention 1 × richness
- Population 1 × intervention 2 × abundance

– Same population (i.e. same taxonomic group), **different** interventions, and **different** measurements.

– **Two unique row**s in the database



Wrangling complex data

Possible formats

condensed

		sample size			
study	site	2000	2007	2010	measured outcomes
Hadley and Burns 20014	Shrewsbury	34	36		bird diversity, tree diversity, vegetation cover
Smith et al. 2012	Bangor			79	coleoptera body mass, tree diversity
Wagner et al. 2015	Leeds	12		15	bird diversity, tree diversity

Complex wrangling required

Complex wrangling required

wide

study	site	sample.size.2000	sample.size.2007	sample.size.2010	outcome.bird.biodiversity	outcome.coleoptera.body.mass	outcome.tree.diversity	outcome.vegetation.cover
Hadley and Burns 20014	Shrewsbury	34	36		1		1	1
Smith et al. 2012	Bangor			79		1	1	
Wagner et al. 2015	Leeds	12		15	1		1	

long

study	site	year	sample.size	outcome
Hadley and Burns 20014	Shrewsbury	2000	34	bird biodiversity
Hadley and Burns 20014	Shrewsbury	2000	34	vegetation cover
Hadley and Burns 20014	Shrewsbury	2000	34	tree diversity
Hadley and Burns 20014	Shrewsbury	2007	36	bird biodiversity
Hadley and Burns 20014	Shrewsbury	2007	36	vegetation cover
Hadley and Burns 20014	Shrewsbury	2007	36	tree diversity
Smith et al. 2012	Bangor	2010	79	coleoptera body mass
Smith et al. 2012	Bangor	2010	79	tree diversity
Wagner et al. 2015	Leeds	2000	12	tree diversity
Wagner et al. 2015	Leeds	2000	12	bird diversity
Wagner et al. 2015	Leeds	2010	15	tree diversity
Wagner et al. 2015	Leeds	2010	15	bird diversity

Wrangling readily feasible in *tidy*

Haddaway et al 2021. *Environ Evid*

Option 1: condensed database

		sample size			
study	site	2000	2007	2010	measured outcomes
Hadley and Burns 20014	Shrewsbury	34	36		bird diversity, tree diversity, vegetation cover
Smith et al. 2012	Bangor			79	coleoptera body mass, tree diversity
Wagner et al. 2015	Leeds	12		15	bird diversity, tree diversity

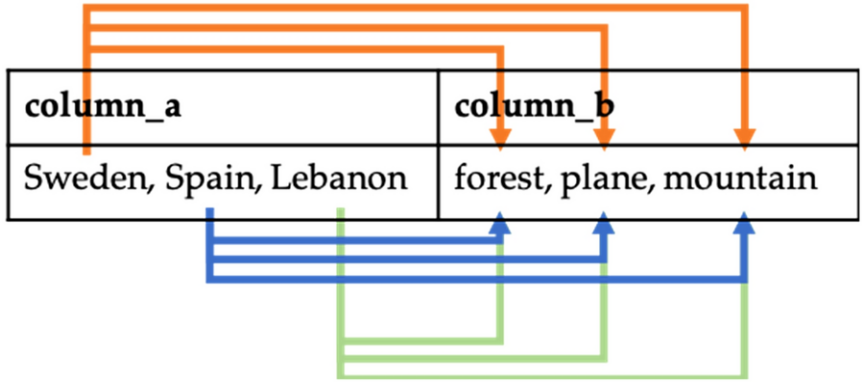
Haddaway et al 2021. *Environ Evid*

1 line = 1 article:

- If **>1 measured outcom**e? They are coded in the same cell.

Avantages :
- Easy to read.
- Easy to code

Disadvantages:
- If several outcomes: need to wrangle data for data analysis.
- **No link** can be made between *Sample size* et *Outcome* => **« linkage loss »**



Haddaway et al 2021. *Environ Evid*

Option 2: wide database

study	site	sample.size.2000	sample.size.2007	sample.size.2010	outcome.bird.biodiversity	outcome.coleoptera.body.mass	outcome.tree.diversity	outcome.vegetation.cover
Hadley and Burns 20014	Shrewsbury	34	36		1		1	1
Smith et al. 2012	Bangor			79		1	1	
Wagner et al. 2015	Leeds	12		15	1		1	

Haddaway et al 2021. *Environ Evid*

1 line = 1 article:

- If **>1 measured outcome?** Outcomes are separated by columns **(1 = yes : binary coding)**

Advantages:

- Much easier to filter.
- Easier to synthesise.
- Columns can be combined/linked.

Disadvantages:

- **But again,** we **can lose** some information**.** In this example, **no link** can be made between *Sample size* et *Outcome* => **« linkage loss »**

Option 3: Long

study	site	year	sample.size	outcome
Hadley and Burns 20014	Shrewsbury	2000	34	bird biodiversity
Hadley and Burns 20014	Shrewsbury	2000	34	vegetation cover
Hadley and Burns 20014	Shrewsbury	2000	34	tree diversity
Hadley and Burns 20014	Shrewsbury	2007	36	bird biodiversity
Hadley and Burns 20014	Shrewsbury	2007	36	vegetation cover
Hadley and Burns 20014	Shrewsbury	2007	36	tree diversity
Smith et al. 2012	Bangor	2010	79	coleoptera body mass
Smith et al. 2012	Bangor	2010	79	tree diversity
Wagner et al. 2015	Leeds	2000	12	tree diversity
Wagner et al. 2015	Leeds	2000	12	bird diversity
Wagner et al. 2015	Leeds	2010	15	tree diversity

Haddaway et al 2021. *Environ Evid*

1 line = 1 outcome:

- One line by ***comparative study*.**

Advantages :

- Ideal for filtering and synthesising.
- Can link sample sizes to outcomes (no linkage loss)
- **Best for meta-analyses**

Disadvantages:

- Difficult to read.
- The same article/study can be repeated several times **(i.e. separated into « study units »).**
- Need to devise a **study code** and **article code**

Consistency checking

Training set

What is training?

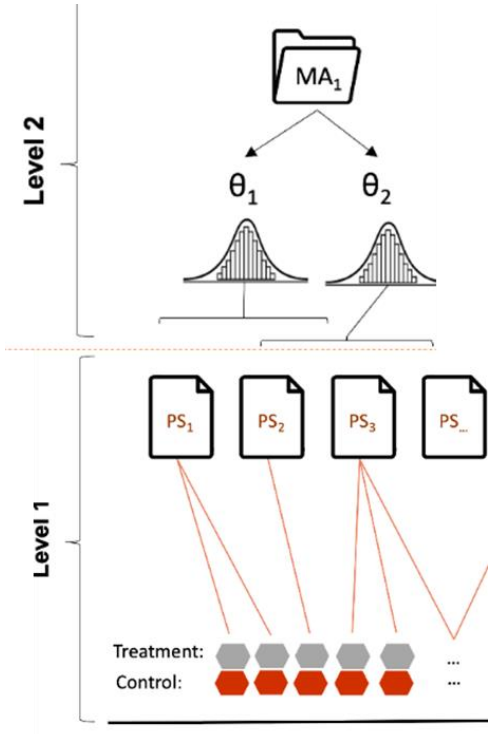
Data extraction of each study carried out **independently** by **2 people**.

Then,

- If several people are involved, conduct a **formal pilot test** on a sample of the data to **check consistency** between reviewers before starting the full data extraction, and discuss any disagreements.

- If only one person is involved, ask another person to **check a sample of the extracted data** at the beginning of the process, and **discuss any disagreements.**

Hierarchical data structure



After Bonfanti et al. 2025. *Ecology letters*

Level 2: the **meta-analysis:** calculation of an overall average (i.e. aggregating the means of each *comparative study* within each individual *primary study*)

Level 1: individual *primary studies*



Comparative study or **paired data**: the individual primary studies report one or several paired data comparisons on an outcome of interest

Practical session
and working example

Working example: structure of a *long* database

Article_Id	Biblio.ID	Biblio.Year	Comp.num	Comp.code	DIV.IND	M.DIV.UNM	SD.DIV.UNM	N.DIV.UNM	NBIS.DIV.UN	M.DIV.MAN	SD.DIV.MAN	N.DIV.MAN	NBIS.DIV.MA
28	28	201	1	Orczewska_2019_1	SR	28.9	6.37	28	11	32.52	9.15	30	11
28	28	201	2	Orczewska_2019_2	SR	4.63	2.37	28	11	9.78	5.16	30	11
28	28	201	3	Orczewska_2019_3	SR	16.85	3.92	28	11	15.61	3.1	30	11
35	35	2019	1	Leski_2019_1	SR	11.41	6.33	4	1	20.79	5.82	4	1
35	35	2019	2	Leski_2019_2	SR	16.07	5.66	4	1	23.02	4.27	4	1
35	35	2019	3	Leski_2019_3	SR	26.84	6.09	4	1	26.59	8.31	4	1
35	35	2019	4	Leski_2019_4	SR	20.11	3.32	4	1	26.57	5.92	4	1
35	35	2019	5	Leski_2019_5	SR	32.45	2.82	4	1	32.43	3.4	4	1
35	35	2019	6	Leski_2019_6	SR	38.92	4.24	4	1	37.06	3.47	4	1
59	59	2019	1	Suominen_2019_1	SR	17.57	7.36	3	3	25.1	6.69	3	3
59	59	2019	2	Suominen_2019_2	SR	17.57	7.36	3	3	22.59	9.93	6	6

A Comparative study = a taxon group × an intervention × an outcome (SR, Abun)
= e.g. **lichens** × **stand not managed since at least 20 years** × **mean species richness**

Comparative study level: ex. several possible comparators (specific extraction of timber)

Aligning data extraction with PICO elements

P I C O : **population** elements

Comparison code	Study Population: Kingdom	Study Population: branch	Study Population: Class	Study Population: Order	Study Population: Family	Taxonomic subset	Study Population: Ecological classification
Orczewska_2019_1	Plants	Vascular plants	NA	NA	NA	Total	Herbs
Orczewska_2019_2	Plants	Vascular plants	NA	NA	NA	Subset	Open-habitat species
Orczewska_2019_3	Plants	Vascular plants	NA	NA	NA	Subset	Ancient forest species

P I **C** O : **comparator** elements

MAN: crit.class	Management type	Management type 1	Management type 2	Management type 3
Clear cutting (cutting all the trees)	Salvage logging	1a	2aa	3b
Clear cutting (cutting all the trees)	Salvage logging	1a	2aa	3b
Clear cutting (cutting all the trees)	Salvage logging	1a	2aa	3b

Aligning data extraction with PICO elements

P I C **O** : **outcome** elements

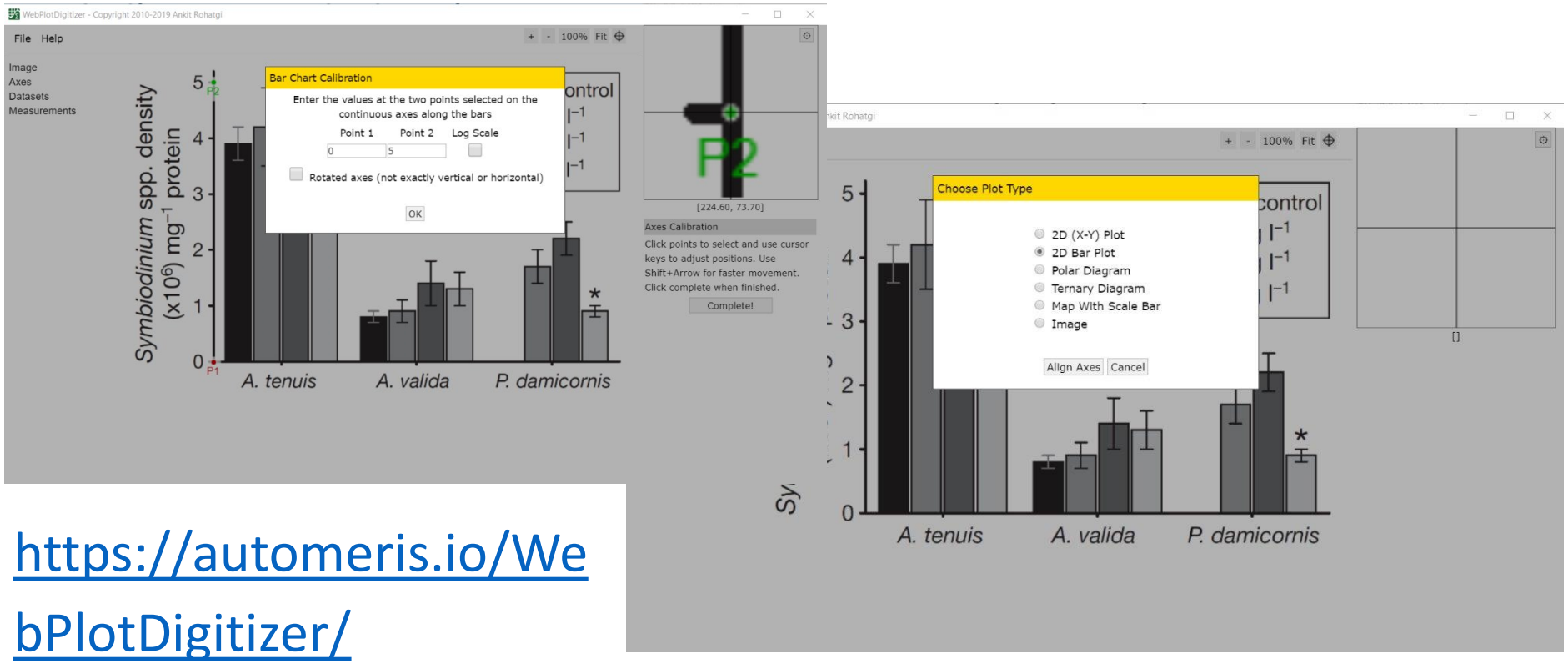
Richness : Indice	MeanRichness: UnManaged	SD_SR : UnManaged	N_SR : UnManaged	MeanRichness: Managed	SD_SR : Managed	N_SR : Managed	Nbis_SR : Managed
SR	28,90	6,37	28	32,52	9,15	30	11
SR	4,63	2,37	28	9,78	5,16	30	11
SR	16,85	3,92	28	15,61	3,10	30	11

Top tip !
Code your data source

Data Comments
Data extract from boxplot FIG 4
Data extract from boxplot FIG 4
Data extract from boxplot FIG 4

Tools for extracting data

webplotdigitizer



https://automeris.io/WebPlotDigitizer/

Tools for extracting data

metaDigitise

Received: 13 July 2018 | Accepted: 12 October 2018
DOI: 10.1111/2041-210X.13118

APPLICATION

Methods in Ecology and Evolution BRITISH ECOLOGICAL SOCIETY

Reproducible, flexible and high-throughput data extraction from primary literature: The metaDigitise r package

Joel L. Pick | Shinichi Nakagawa | Daniel W. A. Noble

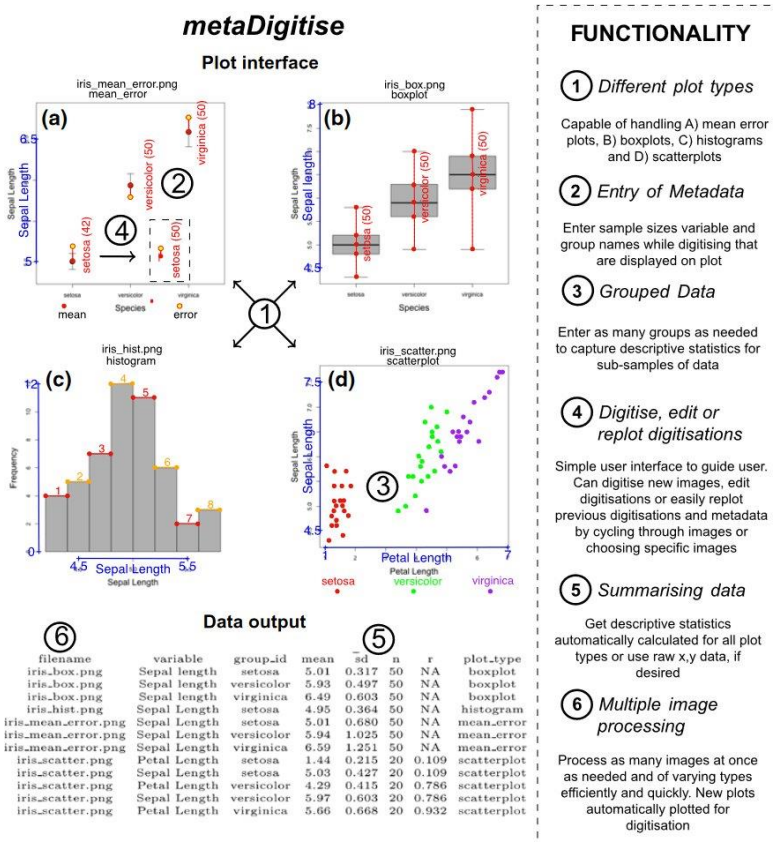
Advantages: possible to save, trace and modify data extraction

Disadvantages: no zoom

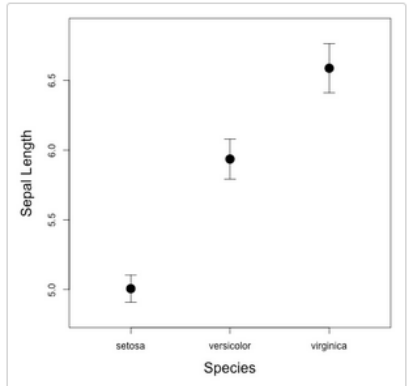
https://cran.r-project.org/web/packages/metaDigitise/vignettes/metaDigitise.html

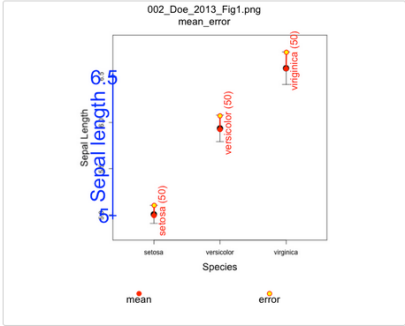
Tools for extracting data

metaDigitise



filename	variable	group_id	mean	sd	n	r	plot_type
iris_box.png	Sepal length	setosa	5.01	0.317	50	NA	boxplot
iris_box.png	Sepal length	versicolor	5.93	0.497	50	NA	boxplot
iris_box.png	Sepal length	virginica	6.49	0.603	50	NA	boxplot
iris_hist.png	Sepal Length	setosa	4.95	0.364	50	NA	histogram
iris_mean_error.png	Sepal Length	setosa	5.01	0.680	50	NA	mean_error
iris_mean_error.png	Sepal Length	versicolor	5.94	1.025	50	NA	mean_error
iris_mean_error.png	Sepal Length	virginica	6.59	1.251	50	NA	mean_error
iris_scatter.png	Petal Length	setosa	1.44	0.215	20	0.109	scatterplot
iris_scatter.png	Sepal Length	setosa	5.03	0.427	20	0.109	scatterplot
iris_scatter.png	Petal Length	versicolor	4.29	0.415	20	0.786	scatterplot
iris_scatter.png	Sepal Length	versicolor	5.97	0.603	20	0.786	scatterplot
iris_scatter.png	Petal Length	virginica	5.66	0.668	20	0.932	scatterplot





! Attention ! sometimes Rstudio buggs, use R consol

Practical session

metaDigitise

Question: estimate the effect of abandonment of forest management on species richness of multiple taxa

Sample of 3 articles:

A **comparative study** = a taxon × an intervention × an outcome

Practical session

metaDigitise

Extracting data

1. Install et load metaDigitise package; set working directory
2. Prepare the figure files (screen capture .png), put them in a "figs" folder in the working directory
3. Run metadigistise > dat <- metaDigitise(dir = "./figs")
4. Use package to extract data into an Excel file
5. If you need to convert a se to sd cf. excel file :

Discussion

Practical session

metaDigitise

```
install.packages("metaDigitise")

library(metaDigitise)

install.packages("writexl")

library(writexl)


extract_dat <- metaDigitise("C:/Users/33676/Documents/blahblahala/Metadigitise")
write_xlsx(extract_dat, "metadigitise_output.xlsx")
```



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